

Note on the Exceptional Set in the ABC Conjecture

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Abstract

Fix $\varepsilon > 0$, let $x > 1$ be a large real number and let $\text{rad}(n) = \prod_{p|n} p$ be the radical of an integer $n \geq 1$. A triple (a, b, c) , with $a+b=c$ and $\gcd(a, b, c) = 1$, such that $c > (\text{rad}(abc))^{1+\varepsilon}$, is called exceptional triple. Recent works have proved that the cardinality $\#\mathcal{E}(x)$ of set $\mathcal{E}$ of exceptional triples satisfies $\#\mathcal{E}(x) = O(x^{2/3})$. This note proves that the cardinality of the exceptional set $\mathcal{E}(x)$ of triples (a, b, c) is an infinite set unconditionally.

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1 Introduction

The symbol $\mathbb{N} = \{1, 2, 3, \dots\}$ denotes the set of natural numbers. Let $\varepsilon > 0$ be a fixed small real number, let $x > 1$ be a large real number and let $\text{rad}(n) = \prod_{p|n} p$ be the radical of an integer $n \geq 1$. The integer solutions $(a, b, c) \in \mathbb{N} \times \mathbb{N} \times \mathbb{N}$ of the equation $X + Y = Z$, with $\gcd(a, b, c) = 1$, such that

$$c > \text{rad}(abc)^{1+\varepsilon}, \quad (1.1)$$

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are called exceptional triples. The set of exceptional triples is defined by

$$\mathcal{E}(x) = \left\{ (a, b, c) \in \mathbb{N}^3 : \gcd(a, b, c) = 1; a + b = c; \right. \\ \left. c > x; \text{ and } c > \text{rad}(abc)^{1+\varepsilon} \right\}. \quad (1.2)$$

The exceptional triples are closely related to counterexamples of the *abc* conjecture.

Conjecture 1.1. (The *abc* conjecture) Given a fixed real number $\varepsilon > 0$, there exists a constant c_ε such that if $a + b = c$ and $\gcd(a, b, c) = 1$, then

$$c \leq c_\varepsilon \text{rad}(abc)^{1+\varepsilon}. \quad (1.3)$$

There are several formulations of the *abc* conjecture over the integers, as well as generalizations to number fields and function fields, see [3], [6] and the vast literature.

It was recently proved that the cardinality of set $\mathcal{E}$ of exceptional triples satisfies $\#\mathcal{E}(x) = O(x^{2/3})$, see [1] and [5]. This note provides a qualitative result about the exceptional set.

Theorem 1.1. *For any small real number $\varepsilon > 0$ and large $x \geq x_0$, the followings hold.*

- (i) $\mathcal{E}(x) \neq \emptyset$.
- (ii) $\#\mathcal{E}(x) \rightarrow \infty$ as $x \rightarrow \infty$.

In particular, the union of exceptional sets

$$\mathcal{E} = \bigcup_{x \geq x_0} \mathcal{E}(x) \quad (1.4)$$

is an infinite subset of $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$ and in the limit, the cardinality of the exceptional set

$$\#\mathcal{E} = \lim_{x \geq x_0} \#\mathcal{E}(x) \quad (1.5)$$

is unbounded.

The proof is derived from the theory of smooth integers in short intervals. The fundamental background is covered in Section 3 to Section ???. The main result is proved in Section 5 and Section 6.

2 Basic Theory of Smooth Numbers

A few results in the theory of smooth numbers are stated in this section as a reference. The symbol $P(n)$ represents the largest prime divisor of an integer $n \geq 1$ with the convention $P(1) = 1$. For $x > y > 1$, the parameter $u = \log x / \log y$. For

$1 < y < x$, an integer $n \geq 1$ is called y -smooth if all its prime factors $p \mid n$ satisfy $p \leq y$. Define the set of smooth integers by

$$S(x, y) = \{n \leq x : P(n) \leq y\} \quad (2.1)$$

and its complementary set

$$\overline{S}(x, y) = \{n \leq x : P(n) > y\}. \quad (2.2)$$

The corresponding counting functions are defined by

$$\Psi(x, y) = \#\{n \leq x : P(n) \leq y\} \quad (2.3)$$

and its complementary set

$$\Phi(x, y) = \#\{n \leq x : P(n) > y\}. \quad (2.4)$$

Definition 2.1. The Dickman function $\rho : [0, \infty] \rightarrow [0, 1]$ is a unique continuous function such that

$$\rho(u) = \begin{cases} 1 & \text{if } 0 \leq u \leq 1, \\ 1 - \log u & \text{if } 1 \leq u \leq 2 \end{cases} \quad (2.5)$$

and satisfies the differential equation

$$u\rho'(u) - \rho(u-1) = 0, \quad (2.6)$$

for $u \geq 1$.

As usual, the parameter u is defined by

$$u = \frac{\log x}{\log y}, \quad (2.7)$$

and the cardinalities of the sets $S(x, y)$ and its complement $\overline{S}(x, y)$ have the asymptotics

$$\Psi(x, y) = \Psi(x, x^{1/u}) \sim x\rho(u), \quad (2.8)$$

and

$$\Phi(x, y) = \Phi(x, x^{1/u}) \sim x(1 - \rho(u)). \quad (2.9)$$

Other properties of interest in this analysis are listed here.

Theorem 2.1. *The function $\rho(u)$ satisfies the followings.*

- (i) $\rho(u) > 0$, for $u > 0$,
 - (ii) $u\rho(u) = \int_{u-1}^u \rho(t)dt$, for $u \geq 1$,
 - (iii) $\rho(u) \leq \frac{1}{\Gamma(u+1)}$, for $u \geq 0$,
-

- (iv) $\rho(u) = e^{-u(\log u + \log \log u - 1 + o(1))}$, for $u \geq 1$,
 (v) $\rho'(u) < 0$, for $u > 1$,

An elementary introduction to smooth numbers appears in [2, Chapter 9].
 The smooth numbers counting function has the asymptotic formula

Theorem 2.2. *For any fixed $\varepsilon > 0$ the relation*

$$\Psi(x, y) = x\rho(u) \left(1 + O\left(\frac{\log(u+1)}{\log y} \right) \right) \quad (2.10)$$

holds uniformly (with $u = \log x / \log y$) in the range

$$y \geq 2 \quad \text{and} \quad 1 \leq u \leq (\log y)^{3/5}. \quad (2.11)$$

Proof. The result is [4, Theorem 1.1], a slightly different form of this asymptotic formula appears in [2, Theorem 9.14]. ■

Surprisingly, this result proved in 1993 seems to imply the latest results for short intervals discussed in Section 3.

3 Smooth Integers in Short Intervals

The topic of smooth integers in short intervals is a major area of research in the theory of smooth integers.

Theorem 3.1. *Let $17/30 < \theta \leq 1$. There exists a constant $C = C(\theta) > 0$ such that the estimate*

$$\frac{\Psi(x+h, y) - \Psi(x, y)}{h} = \frac{\Psi(x, y)}{x} \left(1 + O_\theta \left(\frac{\log(u+1)}{\log y} \right) \right) \quad (3.1)$$

holds uniformly for

$$x^\theta \leq h \leq x \quad \text{and} \quad e^{C(\log x)^{2/3}(\log \log x)^{4/3}} \leq y \leq 2x. \quad (3.2)$$

Proof. The result is [9, Theorem 1.1]. ■

Another result for shorter intervals is stated here.

Theorem 3.2. *For any $\varepsilon > 0$, there exists a positive constant $c_0 = c_0(\varepsilon)$ such that the following holds. If x is large in terms of ε ,*

$$\exp(c_0(\log x)^{2/3}(\log \log x)^{4/3}) \leq y \leq x^{\frac{1}{c_0}}, \quad (3.3)$$

and

$$h \geq \sqrt{x} \exp \left((1 + \varepsilon) \left(\frac{11}{16} u \log u + 2 \log \log x \right) \right), \quad (3.4)$$

where $u = \frac{\log x}{\log y}$, then the interval $[x, x+h]$ contains a y -smooth number.

Proof. The result is [7, Theorem 1.2]. ■

Corollary 3.1. *Let $\theta = 3/5$ and let $x > 1$ be a sufficiently large real number. Set the parameters*

$$B = e^{c_0(\log x)^{2/3}(\log \log x)^{4/3}} \quad (3.5)$$

$$h = x^{3/5}, \quad (3.6)$$

where $c_0 = c_0(\theta) > 0$ is a constant. Then the short interval

$$[x, x + h] \quad (3.7)$$

contains

$$\Psi(x + h, B) - \Psi(x, B) = x^{3/5} e^{-(\log x)^{1/3+\beta}} (1 + o(1)) \quad (3.8)$$

B -smooth integers, for some small number $\beta > 0$.

Proof. The parameters

$$B = y = \exp \left(c_0(\log x)^{2/3}(\log \log x)^{4/3} \right), \quad (3.9)$$

$$h = x^{3/5} > x^{17/30}, \quad (3.10)$$

$$u = \frac{\log x}{\log y} = \frac{\log x}{c_0(\log x)^{2/3}(\log \log x)^{4/3}} = \frac{(\log x)^{1/3}}{c_0(\log \log x)^{4/3}}, \quad (3.11)$$

$$\rho(u) = e^{-u(\log u + \log \log u - 1 + o(1))} = e^{-(\log x)^{1/3+\beta}}, \quad (3.12)$$

lie in the admissible range of Theorem 3.1. The properties of Dickman function $\rho(u)$ are described in Section 2. Thus, the short interval

$$[x, x + h] = [x, x + x^{3/5}] \quad (3.13)$$

contains

$$\begin{aligned} \frac{\Psi(x + h, y) - \Psi(x, y)}{h} &= \frac{\Psi(x, y)}{x} \left(1 + O_\theta \left(\frac{\log(u + 1)}{\log y} \right) \right) \\ &= \rho(u) \left(1 + O_\theta \left(\frac{1}{(\log x)^{2/3}(\log \log x)^{1/3}} \right) \right) \end{aligned} \quad (3.14)$$

B -smooth integers. The exponentially large short interval $[x, x + x^{3/5}]$ contains a substantial number of B -smooth integers. Specifically, (3.14) reduces to

$$\begin{aligned} \Psi(x + h, B) - \Psi(x, B) &= x^{3/5} \rho(u) \left(1 + O_\theta \left(\frac{1}{(\log x)^{2/3}(\log \log x)^{1/3}} \right) \right) \\ &= x^{3/5} e^{-(\log x)^{1/3+\beta}} (1 + o(1)). \end{aligned} \quad (3.15)$$

This completes the proof. ■

Observe that

$$\rho(u) = e^{-\frac{(\log x)^{1/3}}{3c_0(\log \log x)^{4/3}}(1+o(1))} = e^{-(\log x)^{1/3+\beta}} \rightarrow 0 \text{ as } x \rightarrow \infty, \quad (3.16)$$

but the short interval has exponential size.

Lemma 3.1. *Let $\theta = 3/5$ and let $x > 1$ be a large real number. Set*

$$B = e^{c_0(\log x)^{2/3}(\log \log x)^{4/3}}, \quad (3.17)$$

where $c_0 = c_0(\theta) > 0$ is a suitable constant. Let N be a B -smooth integer such that

$$N \in [x, x + x^{3/5}]. \quad (3.18)$$

Assume that $\omega(N) \leq 2 \log \log x$, then the radical satisfies the inequality

$$\text{rad}(N) = \prod_{p|N} p \leq e^{2c_0(\log x)^{2/3}(\log \log x)^{7/3}}. \quad (3.19)$$

Proof. By hypothesis, the number of prime divisors

$$p_1 < p_2 < \cdots < p_w \quad (3.20)$$

of the smooth integer N is

$$w = \omega(N) = 2 \log \log x. \quad (3.21)$$

Now, by definition of the radical and since every prime divisor p_i of N is at most $p_i \leq B$,

$$\begin{aligned} \text{rad}(N) &= \prod_{p|N} p \\ &\leq p_1 \cdot p_2 \cdots p_w \\ &\leq B^{\omega(N)}. \end{aligned} \quad (3.22)$$

Accordingly, radical satisfies the inequality

$$\begin{aligned} \text{rad}(N) &= \prod_{p|N} p \\ &\leq \left(e^{c_0(\log x)^{2/3}(\log \log x)^{4/3}} \right)^{2 \log \log x} \\ &\leq e^{2c_0(\log x)^{2/3}(\log \log x)^{7/3}}. \end{aligned} \quad (3.23)$$

This completes the verification. ■

4 Density of Subsets of Smooth Integers

A Turan-type argument, as in the proof of [8, Theorem], is utilized here to develop a simple verification that the B -smooth integers on the interval $[x, x^{3/5}]$ with an extreme number of prime factors has zero density on this interval. Similar result can be obtained via the Turan-Kubilus inequality.

Lemma 4.1. *Let $x > 0$ be a large real number and let $y = c_0(\log x)^{2/3}(\log \log x)^{4/3}$, where $c_0 > 0$ is a suitable constant. If $u > u_0 \geq 0$, then*

$$\rho\left(\frac{\log x/p}{\log y}\right) = \rho(u) + O\left(\frac{\log p}{\log y} \cdot \frac{(\log \log x)^8}{(\log x)^2}\right). \quad (4.1)$$

Proof. Set $u = \log x / \log y$. Since $\rho(u)$ is a continuous function of $u \geq 0$, the mean value theorem (or a series expansion) generates the linear approximation

$$\begin{aligned} \rho(u_0) &= \rho\left(\frac{\log x/p}{\log y}\right) \\ &= \rho\left(u - \frac{\log p}{\log y}\right) \\ &= \rho(u) + O\left(\frac{\log p}{\log y} \max_{v \in [u-1, u]} |\rho'(v)|\right) \\ &= \rho(u) + O\left(\frac{\log p}{\log y} \cdot \frac{1}{u^6}\right). \end{aligned} \quad (4.2)$$

The upper bound for $|\rho'(u)|$ is derived from the relation $u\rho'(u) - \rho(u-1) = 0$. More precisely, it implies that

$$\begin{aligned} |\rho'(u)| &= \frac{\rho(u-1)}{u} \\ &\leq \frac{1}{u} \cdot \frac{1}{\Gamma(u)} \\ &\leq \frac{1}{u^6} \end{aligned} \quad (4.3)$$

for large $u > 1$ since $\rho(u) \leq 1$ and $u^5 \leq \Gamma(u)$ for large $u > 1$, see Theorem 2.1 for more details. Replacing the value for u yields

$$\frac{1}{u^6} = \left(\frac{\log y}{\log x}\right)^6 = \left(\frac{c_0(\log x)^{2/3}(\log \log x)^{4/3}}{\log x}\right)^6 = \frac{c_1(\log \log x)^8}{(\log x)^2}, \quad (4.4)$$

where $c_1 = c_0^6$ is a suitable constant. ■

Lemma 4.2. *Let x be a large real number, $h = x^\theta > x^{7/12}$ and $y < h < x$. If $y = o(h)$, then*

$$\sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} \omega(n) = h\rho(u)(\log \log y) \left(1 + O_\theta \left(\frac{\log(u+1)}{\log y} \right) \right) + O(h\rho(u)), \quad (4.5)$$

where $u = \log x / \log y$.

Proof. By definition

$$\sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} \omega(n) = \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} \sum_{p|n} 1 = \sum_{p \leq y} \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y \\ p|n}} 1. \quad (4.6)$$

Utilizing [Theorem 3.1](#) to evaluate the inner sum yields

$$\begin{aligned} \sum_{p \leq y} \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y \\ p|n}} 1 &= \sum_{p \leq y} \left(\Psi \left(\frac{x+h}{p}, y \right) - \Psi \left(\frac{x}{p}, y \right) \right) \\ &= \sum_{p \leq y} \frac{h}{p} \rho(u_0) \left(1 + O_\theta \left(\frac{\log(u_0+1)}{\log y} \right) \right), \end{aligned} \quad (4.7)$$

where $u_0 = \log(x/p) / \log y$. Substituting the linear approximation given in [Lemma 4.1](#) and evaluating the finite sum yield

$$\begin{aligned} \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} \omega(n) &= \sum_{p \leq y} \left(\frac{h}{p} \left(\rho(u) + O \left(\frac{\log p}{\log y} \cdot \frac{1}{u^6} \right) \right) \left(1 + O_\theta \left(\frac{\log(u+1)}{\log y} \right) \right) \right) \\ &= h\rho(u)(\log \log y) \left(1 + O_\theta \left(\frac{\log(u+1)}{\log y} \right) \right) + O(h\rho(u)), \end{aligned} \quad (4.8)$$

where

$$O \left(\frac{h}{\log y} \cdot \frac{1}{u^6} \sum_{p \leq y} \frac{\log p}{p} \right) = O(h). \quad (4.9)$$

since $u^6 > (\log y)^2$ is large, see . ■

Lemma 4.3. *Let x be a large real number, then*

$$A(x) = \sum_{\substack{p, q \leq x \\ p \neq q}} \frac{1}{p} \cdot \frac{1}{q} = (\log \log x)^2 + O(\log \log x). \quad (4.10)$$

Proof. First, consider the identity

$$\sum_{\substack{p, q \leq x \\ p \neq q}} \frac{1}{p} \cdot \frac{1}{q} = \left(\sum_{p \leq x} \frac{1}{p} \right)^2 - \sum_{p \leq x} \frac{1}{p^2}. \quad (4.11)$$

Now, the claim follows from Mertens theorem

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + O(1), \quad (4.12)$$

and the diagonal contribution $\sum_{p \leq x} p^{-2} = O(1)$. ■

Lemma 4.4. *Let x be a large real number, $h = x^\theta > x^{7/12}$ and $y < h < x$. If $y = o(h)$, then*

$$\sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} \omega^2(n) = h\rho(u)(\log \log y)^2 + O(h\rho(u) \log \log y), \quad (4.13)$$

where $u = \log x / \log y$.

Proof. Expanding the square

$$\omega^2(n) = \left(\sum_{p|n} 1 \right)^2 = \sum_{p|n} 1 + \sum_{\substack{p, q|n \\ p \neq q \\ p, q \text{ primes}}} 1, \quad (4.14)$$

and switching the order of summations lead to

$$\begin{aligned} \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} \omega^2(n) &= \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} \left(\sum_{p|n} 1 + \sum_{\substack{p, q|n \\ p \neq q \\ p, q \text{ primes}}} 1 \right) \\ &= \sum_{p \leq y} \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y \\ p|n}} 1 + \sum_{p, q \leq y} \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y \\ p, q|n \\ p \neq q}} 1. \end{aligned} \quad (4.15)$$

The first double finite sum

$$\sum_{p \leq y} \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y \\ p|n}} 1 = h\rho(u)(\log \log y)(1 + o(1)) + O(h\rho(u)), \quad (4.16)$$

see [Lemma 4.2](#). Utilize [Lemma 4.3](#) to evaluate the inner sum in the second double finite sum

$$\begin{aligned} \sum_{p,q \leq y} \sum_{\substack{x \leq n \leq x+h \\ p,q|n \\ p \neq q}} 1 &= \sum_{p,q \leq y} \left(\Psi \left(\frac{x+h}{pq}, y \right) - \Psi \left(\frac{x}{pq}, y \right) \right) \\ &= \sum_{p,q \leq y} \frac{h}{pq} \rho(u_1) \left(1 + O_\theta \left(\frac{\log(u_1 + 1)}{\log y} \right) \right), \end{aligned} \quad (4.17)$$

where $u_1 = \log(x/pq)/\log y$. Substituting the linear approximation given in [Lemma 4.1](#) and evaluating the finite sum yield

$$\begin{aligned} \sum_{p,q \leq y} \sum_{\substack{x \leq n \leq x+h \\ p,q|n \\ p \neq q}} 1 &= h \left(\rho(u) + O \left(\frac{\log pq}{\log y} \cdot \frac{1}{u^6} \right) \right) \left(1 + O_\theta \left(\frac{\log(u_1 + 1)}{\log y} \right) \right) \sum_{p,q \leq y} \frac{1}{pq} \\ &= h \rho(u) \left(1 + O_\theta \left(\frac{\log(u_1 + 1)}{\log y} \right) \right) \\ &\quad \times ((\log \log y)^2 + O(\log \log y)) \\ &= h \rho(u) (\log \log y)^2 + O(h \rho(u) \log \log y), \end{aligned} \quad (4.18)$$

where

$$O \left(\frac{h}{\log y} \cdot \frac{1}{u^6} \sum_{p,q \leq y} \frac{\log pq}{pq} \right) = O(h). \quad (4.19)$$

since $1/u^6 > 1/(\log y)^2$, see [\(4.4\)](#). Summing [\(4.16\)](#) and [\(4.18\)](#) completes the verification. $\blacksquare$

Theorem 4.1. *Let x be a large real number, $h = x^\theta > x^{7/12}$ and $y < h < x$ with $y = o(h)$. If $\delta > 0$, $A > 0$, then the subset of y -smooth integers $n \in [x, x+h]$, for which the inequality*

$$\omega(n) > \log \log y + A(\log \log y)^{1/2+\delta} \quad (4.20)$$

holds has zero density in $[x, x+h]$. In particular, its cardinality is $o(h)$.

Proof. The absolute squared error is

$$\begin{aligned}
R(x, h) &= \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} (\omega(n) - \log \log y)^2 \\
&= \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} \omega^2(n) - 2(\log \log y) \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} \omega(n) + (\log \log y)^2 \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y}} 1 \\
&= h\rho(u)(\log \log y)^2 + O(h\rho(u) \log \log y) \\
&\quad - 2(\log \log y)[h\rho(u)(\log \log y) \left(1 + O_\theta\left(\frac{\log(u+1)}{\log y}\right)\right) + O(h\rho(u))] \\
&\quad + (\log \log y)^2 \left(h\rho(u) \left(1 + O_\theta\left(\frac{\log(u+1)}{\log y}\right)\right)\right) \\
&= O(h\rho(u) \log \log y),
\end{aligned} \tag{4.21}$$

since

$$\frac{\log(u+1)}{\log y} = \frac{\log \log x}{(\log x)^{2/3}}. \tag{4.22}$$

Let $\mathcal{K} = \{n : \omega(n) \geq \log \log y + A(\log \log y)^{1/2+\delta}\}$, let $\#\mathcal{K}$ be its cardinality and set $k \geq A(\log \log y)^{1/2+\delta}$. Then

$$\begin{aligned}
O(h\rho(u) \log \log y) &= \sum_{\substack{x \leq n \leq x+h \\ P(n) \leq y \\ \omega(n) \geq k}} (\omega(n) - \log \log y)^2 \\
&= \sum_{n \in \mathcal{K}} k^2 \\
&\geq k^2 \#\mathcal{K}.
\end{aligned} \tag{4.23}$$

This implies that

$$\#\mathcal{K} = O\left(\frac{h\rho(u) \log \log y}{k^2}\right) = O\left(\frac{h\rho(u)}{(\log \log y)^{2\delta}}\right). \tag{4.24}$$

For the parameters

$$y = \exp \left(c_0 (\log x)^{2/3} (\log \log x)^{4/3} \right), \quad (4.25)$$

$$h = x^{3/5} > x^{17/30}, \quad (4.26)$$

$$u = \frac{\log x}{\log y} = \frac{\log x}{c_0 (\log x)^{2/3} (\log \log x)^{4/3}} = \frac{(\log x)^{1/3}}{c_0 (\log \log x)^{4/3}}, \quad (4.27)$$

$$\rho(u) = e^{-u(\log u + \log \log u - 1 + o(1))} = e^{-(\log x)^{1/3+\beta}}, \quad (4.28)$$

$$\log \log y = \log c_0 (\log x)^{2/3} (\log \log x)^{4/3} \asymp \log \log x, \quad (4.29)$$

the number $\#\mathcal{K}$ -smooth integers with $\omega(n) \geq \log \log x + A(\log \log x)^{1/2+\delta}$ prime factor has the upper bound

$$\#\mathcal{K} = O \left(\frac{h\rho(u)}{(\log \log y)^{2\delta}} \right) = O \left(\frac{x^{3/5} e^{-(\log x)^{1/3+\beta}}}{(\log \log x)^{2\delta}} \right), \quad (4.30)$$

for some suitable constants $c_0 = c_0(\theta) > 0$ and for some small real numbers $\beta > 0$, $\delta > 0$. ■

5 Exceptional Triples in Short Intervals

This counterexample is based on the established theory of smooth integers in short intervals. The notations $\text{rad}(abc) < c^{1-\varepsilon_0}$ and $c > \text{rad}(abc)^{1+\varepsilon}$ are equivalent, but $\varepsilon_0 = \varepsilon/(1+\varepsilon)$. So the proof can be written in either form.

Theorem 5.1. *Let $\theta = 3/5$ and let $x > 1$ be a sufficiently large real number. Suppose that*

- | | |
|---|--|
| (i) $B = e^{c_0 (\log x)^{2/3} (\log \log x)^{4/3}},$ | <i>Smooth parameter.</i> |
| (ii) $h = x^{3/5},$ | <i>Short interval size.</i> |
| (iii) $\#\{n \leq x : \omega(n) > w\} = o(x^{3/5}),$ | <i>Most smooth integers in $[x, x+h]$ have $\omega(n) \leq w$ prime factors.</i> |

where $c_0 = c_0(\theta) > 0$ is a suitable constant and $w = A(\log \log y)^{1/2+\delta}$, $A > 0$ constant. Then, as $x \rightarrow \infty$, there are infinitely many B -smooth integers

$$c \in [x, x + x^{3/5}], \quad (5.1)$$

such that $a + b = c$ and the abc inequality fails:

$$c \leq c_\varepsilon (\text{rad}(abc))^{1+\varepsilon}, \quad (5.2)$$

where $\gcd(a, b, c) = 1$.

Proof. Fix a small number $\varepsilon > 0$. Let $x = p^k$ be a large prime power, where $B < p$ and $k > 3$ such that

$$\left(\frac{3}{5} + \frac{1}{k}\right)(1 + \varepsilon) < 1. \quad (5.3)$$

By [Corollary 3.1](#) there exists a B -smooth integer in the short interval

$$c = p_1^{e_1} \cdot p_2^{e_2} \cdots \cdot p_w^{e_w} \in [x, x + h], \quad (5.4)$$

where $p_i \leq B$ are distinct prime, $e_i \geq 0$. And $w < \log \log y + A(\log \log y)^{1/2+\delta} \leq 2 \log \log x$ follows from [Theorem 4.1](#). Now, let

$$\begin{aligned} a &= |p^k - p_1^{e_1} \cdot p_2^{e_2} \cdots \cdot p_w^{e_w}| \leq h \\ b &= p^k \\ c &= p_1^{e_1} \cdot p_2^{e_2} \cdots \cdot p_w^{e_w}. \end{aligned} \quad (5.5)$$

A short inspection shows that, $\gcd(a, b, c) = 1$, see [Remark 5.1](#). Next, take the abc inequality to derive a contradiction:

$$\begin{aligned} c &\leq c_\varepsilon (\text{rad}(abc))^{1+\varepsilon} \\ &\leq c_\varepsilon (\text{rad}(h \cdot p^k \cdot p_1 \cdot p_2 \cdots \cdot p_w))^{1+\varepsilon} \\ &\leq c_\varepsilon \left(x^{\frac{3}{5}} \cdot x^{\frac{1}{k}} \cdot e^{2c_0(\log x)^{\frac{2}{3}}(\log \log x)^{\frac{7}{3}}} \right)^{1+\varepsilon} \\ &\leq c_\varepsilon \left(x^{\frac{3}{5} + \frac{1}{k}} \cdot e^{2c_0(\log x)^{\frac{2}{3}}(\log \log x)^{\frac{7}{3}}} \right)^{(1+\varepsilon)}, \end{aligned} \quad (5.6)$$

where $c_0 > 0$ is a suitable constant. The third line in (5.6) follows from [Lemma 3.1](#). Now, since $k > 3$

$$\begin{aligned} c = x + O(x^{\frac{3}{5}+\delta}) &\leq c_\varepsilon \left(x^{\frac{3}{5} + \frac{1}{k}} \cdot e^{2c_0(\log x)^{\frac{2}{3}}(\log \log x)^{\frac{7}{3}}} \right)^{(1+\varepsilon)} \\ &\leq c_\varepsilon \cdot x^{\left(\frac{3}{5} + \frac{1}{k}\right)(1+\varepsilon)} \cdot e^{2c_0(1+\varepsilon)(\log x)^{\frac{2}{3}}(\log \log x)^{\frac{7}{3}}} \end{aligned} \quad (5.7)$$

is a contradiction infinitely often as $x \rightarrow \infty$. ■

The relationship between k and ε expressed in (5.3) is tabulated in [Table 1](#).

k	4	10	100	1000	∞
$\varepsilon <$	0.176471	0.428571	0.639344	0.663893	.666667

Table 1: The Parameter ε as a Function k

The quality index of the exceptional triples is

$$\begin{aligned}
\gamma_\varepsilon(abc) &= \frac{\log c}{\log \text{rad } abc} \\
&= \frac{\log x + \log(1 + o(1))}{(1 + \varepsilon) \left(\frac{3}{5} + \frac{1}{k} + o(1) \right) \log x + \log c_\varepsilon} \\
&= \frac{1}{(1 + \varepsilon) \left(\frac{3}{5} + \frac{1}{k} \right)} \tag{5.8}
\end{aligned}$$

for large $x > 1$. Fix $\varepsilon = 0.01$, then

k	4	10	100	1000	∞
$\gamma_k(abc)$	1.164822	1.414427	1.623113	1.647419	1.650165

Table 2: Quality Index $\gamma_k(abc)$ as a Function k

As a function of $k > 3$, the quality index of the exceptional triples generated here has a limit $\gamma_k(abc) = 1.650165$ that exceeds the best known index $\gamma(abc) = 1.6299$ for the top abc triple $(a, b, c) = (2, 3^{10} \cdot 109, 23^5)$, see [Table 2](#). The determination of an exceptional triple of index $\gamma_{100}(abc) \approx 1.6393$ involves calculations with extremely large integers $x = p^k = b$ to find a B -smooth integer c in the interval $x, x + x^{3/5}$, this problem is quite similar to the Skewes number problem.

The cube $[1, x] \times [1, x] \times [1, x] \subset \mathbb{N} \times \mathbb{N} \times \mathbb{N}$ contains about $6\pi^{-2}x^2$ relatively primes triples (a, b, c) , the result in [\[5\]](#) shows that about $\#\mathcal{E}(x) = O(x^{2/3})$ fail the abc inequality. The goal is to prove that just $\#\mathcal{E}(x) = O(1)$ fail the abc inequality. However, the analysis within seems to show that the cardinality of exceptional set $\mathcal{E}(x)$ is bounded below, specifically $\#\mathcal{E}(x) \gg x^\delta$ as $x \rightarrow \infty$.

Remark 5.1. In more detail, the relatively prime condition is as follows: Since $p > B \geq P(c)$, it follows that $p \nmid c$ and $\gcd(b, c) = 1$. Any prime q dividing both a and b divides $b = p^k$ and therefore $q = p$; but $p \nmid a$ because $a = |p^k - c|$ and $p \nmid c$. Thus $\gcd(a, b) = 1$. Finally, if a prime q divided a, b, c simultaneously then $q \mid b$ and $q \mid c$, contradicting $p \nmid c$ and the fact that all prime divisors of c are $\leq B < p$. Therefore $\gcd(a, b, c) = 1$.

6 A Lower Bound of the Exceptional Set

The qualitative is much simpler and it is presented here.

Proof: [Theorem 1.1](#). For each large real number $x > 1$, [Theorem 5.1](#) proves that exceptional set

$$\mathcal{E}(x) \neq \emptyset. \tag{6.1}$$

There exists at least one triple (a, b, c) that contradicts the abc inequality (5.2). The set $\mathcal{E}(x)$ are monotonically increasing. Hence, an increasing collection of finite sets with infinite union must have unbounded cardinality. the limit

$$\mathcal{E} = \bigcup_{x \rightarrow \infty} \mathcal{E}(x) \tag{6.2}$$

is infinite. In particular, in the limit, the cardinality of the exceptional set

$$\#\mathcal{E} = \lim_{x \rightarrow \infty} \#\mathcal{E}(x) \tag{6.3}$$

is unbounded. ■

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